RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 09th October 2026	**CIE – 1**	**Max. Marks : 50**
Semester : V	**UG**	**Duration : 90 Min**
Course Title: ENTREPRENEURSHIP & INTELLECTUAL PROPERTY RIGHTS		**Course Code : HS351TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	A software company develops an AI-based tool for legal research and seeks to obtain a patent for the invention in India. Examine the patentability of the AI tool under the Indian Patents Act, 1970, with particular reference to the exclusions relating to computer programmes and software. Discuss the major challenges the company may face in obtaining patent protection.	10	4	5
2.	A company has obtained a patent for an innovative electric-vehicle battery management system. A competitor claims that similar technology was disclosed in an earlier patent. At the same time, an investor wants to know the commercial value of the patent. The company is also considering licensing the technology to an automobile manufacturer. Analyse the case and explain how patent searching, valuation, and commercialisation would help the company make an informed decision.	10	4	5
3.	A technology company develops an AI-based healthcare application that uses a mathematical algorithm to analyse patient data, presents the results through a graphical interface, and recommends a known medicine for treating a particular disease. The company seeks a patent for the entire system. Analyse the patentability of the invention under the Indian Patents Act, 1970, and identify the relevant categories of non-patentable inventions that may constitute grounds for refusal.	10	3	5
4.	"Successful entrepreneurs possess a combination of personal, managerial and behavioural qualities." Discuss the important characteristics and traits of successful entrepreneurs, with suitable examples.	10	2	1
5.	A technology-based start-up wants to use Artificial Intelligence, digital platforms, social entrepreneurship, sustainable business practices and innovative funding models to establish its business. Explain the emerging trends represented in this case and analyse their impact on entrepreneurship.	10	2	1

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DEPARTMENT OF
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Date : 13th April 2026	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Entrepreneurship & Intellectual Property Rights		**Course Code : HS351TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Define Certification Trade Mark.	02	2	5
2.	The Indian government allows a local company to produce a patented medicine during a health emergency without the owner's consent. What is this called under patent law?	02	2	5
3.	A company tries to register the word "Laptop" as its trademark for selling computers. Can this mark be registered?	02	2	5
4.	Dr. Meena discovers a new mathematical formula for predicting weather patterns. Can this discovery be patented?	02	2	5
5.	Analyze how the networking skills influence entrepreneurial success.	02	2	1
	Part – B			
1	Critically assess the statement "Entrepreneurs never fail" and explain how failure is an integral part of entrepreneurial growth.	10	2	1
2	Develop a case study of a successful entrepreneurial venture that utilized a unique method for identifying its opportunity, explaining the approach and its results.	10	2	1
3	An international company develops a method to extract active compounds from a plant traditionally used in Ayurveda, claiming higher efficiency and stability. Can this method be patented, and how does it relate to the protection of traditional knowledge?	10	3	5
4	Discuss the concept of patent infringement in detail. Include various types of infringement with relevant case studies to illustrate the practical application of patent law.	10	3	4
5	Explain the concepts of Patent Search and Patent Drafting. Discuss their importance in the process of patent protection.	10	3	5

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DEPARTMENT OF
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Date : 7th Nov 2025	**CIE - I**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Entrepreneurship &Intellectual property Rights		**Course Code : HS351TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Define Eco labelling	2	2	4
2.	List any two characteristics of a successful entrepreneur?	2	2	1
3.	Define Compulsory licensing.	2	2	4
4.	List any two different types of Entrepreneurs.	2	2	1
5.	Rahul, a software engineer, notices that many small businesses struggle with online payments. He develops a mobile app to simplify digital transactions and starts his own company to offer this solution. Based on the above scenario, identify the type of entrepreneur Rahul represents and explain why.	2	3	1
	Part – B			
1	Define a patent and explain its salient features that make it a valuable legal protection for inventors. Analyze the key factors determining the patentability of an invention.	10	3	4
2	Explain the concepts of Registrable and Non-Registrable Trademarks under the Trade Marks Act, 1999. Discuss the procedure for registration of a Trademark under the ACT.	10	3	4
3.	A biotechnology company, Bio Nova Pvt. Ltd., holds a patent for a unique process of producing eco-friendly bio-plastic. A rival firm, Green Poly Industries, begins manufacturing and selling a similar bio-plastic using a slightly modified process without obtaining permission from Bio Nova. Around the same time, Bio Nova plans to expand its patent portfolio and conduct a patent search for new innovations while also seeking investors to commercialize and value its intellectual property assets. Based on the above scenario, analyze the following issues: 1. What constitutes patent infringement in this case, and what legal remedies are available to Bio Nova under the Patents Act, 1970? 2. How should Bio Nova conduct an effective patent search and prepare a patent draft to protect its future innovations? 3. Discuss the process and importance of commercialization and valuation of Bio Nova's intellectual property in attracting investors.	10	4	5

4	Ms. **Ananya Rao**, a young engineer working in a technology firm, is passionate about developing sustainable energy solutions. While her company supports innovation internally, she is uncertain whether to continue as an intrapreneur within the organization or to start her own entrepreneurial venture focused on solar-powered charging systems for rural areas. She discusses her ideas with a team of colleagues who share her vision, and together they plan to explore market opportunities and assess risks. However, her family believes entrepreneurship is too risky and that only people with large capital or business backgrounds can succeed. Based on the above scenario, analyze the following issues: 1. Differentiate between an entrepreneur and an intrapreneur in the context of Ananya's situation. 2. Discuss the emerging trends in entrepreneurship that could influence Ananya's decision to start a new venture. 3. Evaluate the characteristics of a successful entrepreneur that Ananya and her team should develop to ensure business success. 4. Identify and explain the myths about entrepreneurship reflected in the case and how they can be overcome. 5. Explain the role of entrepreneurial teams in fostering innovation and sustaining growth in startups like Ananya's.	10	4	1
5	Explain the importance of entrepreneurship in promoting engineering innovation and fostering economic growth. Discuss how entrepreneurial initiatives among engineers contribute to technological advancement, job creation, and sustainable development, with suitable examples.	10	3	1

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 28th April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Entrepreneurship & Intellectual Property Rights		**Course Code : HS361TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is the primary difference between an Innovative Entrepreneur and an imitative entrepreneur?	1	2	1
2.	Why do Fabian entrepreneurs tend to be less successful in fast-paced industries?	1	2	1
3.	What is the main reason for the failure of many new entrepreneurial ventures?	1	2	1
4.	Why is risk-taking considered an essential characteristic of an entrepreneur?	1	2	1
5.	What emerging trend in entrepreneurship has been accelerated by digital technologies?	1	2	1
6.	What is the primary difference between "entrepreneurship" and "intrapreneurship"?	1	2	1
7.	What is the primary purpose of intellectual property (IP) laws?	1	2	4
8.	If an inventor publishes their invention before filing a patent, what impact does it have on the patentability?	1	2	4
9.	Which of the following can be patented: A new business model, a new type of software algorithm, or a new pharmaceutical drug?	1	2	4
10.	Can a product that is patented in one country be sold without a patent in another country?	1	2	5
	Part – B			
1a.	Define Entrepreneurship and explain its scope in the context of engineering. Discuss the various areas where entrepreneurship plays a key role in engineering industries.	5	2	1
b.	What are the characteristics and traits of successful entrepreneurs? Discuss how these qualities are essential for engineering entrepreneurs to thrive in a competitive and rapidly changing market.	5	2	1

2.	**FitTrack Innovations** was a wearable tech startup launched in 2017, aiming to develop a fitness tracker focused on personalized health insights. Despite having a unique product, the startup struggled with low sales and negative user feedback. The key issues were poor marketing, unclear product positioning, and a lack of customer engagement. After the initial failure, the founders decided to pivot. They conducted detailed market research, redesigned the product to focus on a more specific niche—tracking chronic health conditions—and revamped their marketing strategy. With a clearer value proposition and targeted audience, **FitTrack Innovations** relaunched and gained significant traction in the health tech market. The company later secured venture capital and expanded its product range. a) What do you think was the most critical factor in FitTrack Innovations' failure, and how could the founders have addressed it earlier to avoid the setback? b) What is a pivot in the context of startups, and why did FitTrack Innovations decide to pivot their business model? c) How would you evaluate the effectiveness of the founders' decision to pivot? What risks might they have faced, and what alternatives could they have considered? d) What are the key factors that determine the success or failure of a product launch for a startup? e) How does securing venture capital help a struggling startup, and how did it impact the recovery of FitTrack Innovations?	10	3	2
3.	'SunRay' Innovations is an engineering startup that has developed a cutting-edge solar powered water purification system aimed at providing clean drinking water to rural areas. The technology is innovative and has the potential to improve the quality of life for many communities. However, 'SunRay' Innovations is struggling to secure funding and has limited market reach due to strong competition from larger companies with more resources. a) Discuss how 'SunRay' Innovations demonstrates the key principles of entrepreneurship through its innovation. b) Identify potential entrepreneurial opportunities for 'SunRay' Innovations to overcome its challenges and scale its business. c) How does 'SunRay' Innovations' focus on sustainable technology reflect the broader role of entrepreneurship in addressing social and environmental challenges? d) How can the company effectively manage risks associated with expansion?	10	3	2
4a.	Define Intellectual Property (IP) and explain its importance in the modern economy. Discuss the different types of intellectual property and their significance for businesses.	5	2	4
b.	Provide an overview of the patent application procedure. Discuss the key steps involved in securing a patent and the timelines for the process.	5	2	4

5.	InnovTech, a startup based in India, has developed an innovative device that converts waste plastic into reusable energy. The company has already filed for a patent for their invention. However, after conducting a thorough patent search, they found that a similar invention had already been patented in the United States. InnovTech believes their technology is more efficient and has a better potential for scalability. Now, InnovTech is considering commercializing their invention both locally and globally. They are also exploring the possibility of licensing their patent to larger corporations or transferring the patent rights. a) Discuss the importance of conducting a patent search in the early stages of product development. How does this process help InnovTech in ensuring the uniqueness of their invention? b) Discuss the steps InnovTech should take to protect its invention and ensure its patentability. c) If InnovTech decides to pursue international protection for their invention, what steps must they follow to ensure global patent protection? d) What options does InnovTech have for transferring or commercializing their patent? What legal remedies could they seek if their patent rights are infringed upon?	10	3	5

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**25th November 2024**	**Maximum Marks**	**10 + 50**
Course Code	**HS351TA**	**Duration**	**120 Mins**
Sem	**V Semester**	**CIE – I**	
ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY RIGHTS			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	PART – A			
1	Define Compulsory license under the patent law.	02	1	4
2	List any two characteristics of successful entrepreneurs.	02	2	1
3	Difference between Intrapreneur & Entrepreneur.	02	1	1
4	List any two non-patentable inventions under the Indian patent Act.	02	2	2
5	List any two myths of entrepreneurs.	02	2	1
	PART – B			
1a	A software company develops an AI tool for legal research and applies for a patent in India. Discuss the challenges it might face under Indian patent law.	05	3	3
b	Explain the difference between a product patent and a process patent with an examples.	05	1	4
2	Create a commercialization strategy for a patent related to renewable energy technology.	10	3	3
3	Propose a framework for enhancing Traditional Knowledge protection under international Intellectual Property laws.	10	2	4
4	Create a case study on how perseverance as a characteristic helped a famous entrepreneur achieve success.	10	3	1
5	Explain how entrepreneurship fosters innovation in an economy with one case example.	10	2	1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max	Quiz	04	02	–	04	–	02	02	06	–	–	–
	Marks	Test	20	10	15	15	–	05	10	25	–	–	–

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
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Date	**25th November 2024**	**Maximum Marks**	**10 + 50**
Course Code	**IM254TA**	**Duration**	**120 Mins**
Sem	**V Semester**	**CIE – I**	
ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY RIGHTS			

Note:
1. Answer all the Questions.

PART – A

Sl. No	Questions	M	BT	CO
1	Define Compulsory license under the patent law.	02	1	4
2	List any two characteristics of successful entrepreneurs.	02	2	1
3	Difference between Intrapreneur & Entrepreneur.	02	1	1
4	List any two non-patentable inventions under the Indian patent Act.	02	2	2
5	List any two myths of entrepreneurs.	02	2	1

PART – B

Sl. No	Questions	M	BT	CO
1.a	A software company develops an AI tool for legal research and applies for a patent in India. Discuss the challenges it might face under Indian patent law.	05	3	3
b	Explain the difference between a product patent and a process patent with an examples.	05	1	4
2	Create a commercialization strategy for a patent related to renewable energy technology.	10	3	3
3	Propose a framework for enhancing Traditional Knowledge protection under international Intellectual Property laws.	10	2	4
4	Create a case study on how perseverance as a characteristic helped a famous entrepreneur achieve success.	10	3	1
5	Explain how entrepreneurship fosters innovation in an economy with one case example	10	2	1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Max Marks	Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
		Quiz	04	02	–	04	–	02	02	06	–	–	–
		Test	20	10	15	15	–	05	10	25	–	–	–

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DEPARTMENT OF
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Date : 09th Oct 2026	**CIE – I A**	**Max. Marks : 50**
Semester : V	**UG**	**Duration : 90 Min**
Course Title: Operations Management		**Course Code : IM352IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
1.	With suitable example, Distinguish between Manufacturing and service processes.	10	2	4
2.	Identify nine competitive priorities used in operations strategy, and explain how a consistent pattern of decisions can develop organizational capabilities.	10	2	2
3.	The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Before commercializing the device and adding it to the existing product line, the owner wants to assess its financial feasibility. The estimated variable cost is Rs.7 per unit, while the annual fixed cost is Rs.56,000. i. If the selling price is set at Rs.25 per unit, determine the number of units that must be produced and sold to break even. Use both algebraic and graphic approaches. ii. The forecasted first-year sales are 10,000 units if the selling price is reduced to Rs.15 per unit. Under this pricing strategy, calculate the total contribution to profit during the first year.	10	3	3
4.	A retailer must decide whether to build a small or large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6, respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = $223,000) or to expand (payoff = $270,000). If a small facility is built and demand is low, there is no reason to expand and the payoff is $200,000. If a large facility is built and demand proves to be low, the choice is to do nothing ($40,000) or to stimulate demand through local advertising. The response to advertising may be either modest or sizable, with their probabilities estimated to be 0.3 and 0.7, respectively. If it is modest, the payoff is estimated to be only $20,000; the payoff grows to $220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is $800,000. Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative—building a small facility or building a large facility—has the higher expected payoff?	10	3	3
5.	A manufacturing company wants to introduce a new product with high variety and low production volume. Explain how it should select an appropriate manufacturing process structure and process strategy.	10	3	4

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Date : 06/11/2025	**CIE– I**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Operations Management		**Course Code : IM352IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	When Amazon decides where to locate its fulfillment centers, it is making a _____ level decision in operations management.	1	2	2
2.	Strategies for change include process improvement, process redesign, and process _____.	1	1	1
3.	Strategic fit refers to the alignment between business strategy and _____ strategy.	1	2	2
4.	In manufacturing, the four process structures are job process, batch process, line process, and _____ process.	1	1	1
5.	Decision-making models in operations include deterministic, probabilistic, and _____ models.	1	1	3
6.	The Fourth Industrial Revolution is driven by technologies such as IoT, AI, and _____.	1	1	1
7.	When a factory balances machine utilization and customer delivery times, it is managing the _____ responsiveness trade-off.	1	2	2
8.	Walmart's use of RFID and data analytics to track inventory is an example of using _____ as a driver of supply chain performance.	1	2	3
9.	In a make-to-order process, production begins only after receiving a _____ from the customer.	1	1	1
10.	Zara's ability to rapidly design and distribute new fashions demonstrates the competitive priority of _____.	1	2	2
	Part – B			
1	Define operations management. Explain the key terms involved in the definition of operations management with a schematic model of a production system.	10	2	1
2	With suitable example, distinguish between manufacturing and service processes.	10	2	1

<table>
<tr><td>3</td><td>A retailer must decide whether to build a small or a large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6, respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = $223,000) or to expand (payoff = $270,000). If a small facility is built and demand is low, there is no reason to expand and the payoff is $200,000. If a large facility is built and demand proves to be low, the choice is to do nothing ($40,000) or to stimulate demand through local advertising. The response to advertising may be either modest or sizable, with their probabilities estimated to be 0.3 and 0.7, respectively. If it is modest, the payoff is estimated to be only $20,000; the payoff grows to $220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is $800,000. Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative—building a small facility or building a large facility—has the higher expected payoff?</td><td>16</td><td>4</td><td>3</td></tr>
<tr><td>4</td><td>A manager is deciding whether to build a small or a large facility. Much depends on the future demand that the facility must serve, and demand may be small or large. The manager knows with certainty the payoffs that will result under each alternative, shown in the following payoff table. What is the best alternative for each decision rule: Maximin. Maximax. Laplace. Minimax Regret.
<table>
<tr><td rowspan="2">Alternative</td><td colspan="2">Possible future demand</td></tr>
<tr><td>Low</td><td>High</td></tr>
<tr><td>Small facility</td><td>200</td><td>270</td></tr>
<tr><td>Large facility</td><td>160</td><td>800</td></tr>
<tr><td>Do nothing</td><td>0</td><td>0</td></tr>
</table></td><td>14</td><td>4</td><td>3</td></tr>
</table>

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
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Date	**25th November 2024**	**Maximum Marks**	**10 + 50**
Course Code	**IM352IA**	**Duration**	**120 Min**
Sem	**V**	**CIE – I**	
OPERATIONS MANAGEMENT			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Operations ensure the efficient use of _____ to meet customer demands.	01	1	1
2	A supply chain includes the flow of materials, information, and _____ across organizations.	01	1	1
3	Effective supply chain management ensures timely delivery and reduces _____.	01	1	1
4	Competitive priorities like cost, quality, and _____ determine how a company competes in the market.	01	1	1
5	The Fourth Industrial Revolution integrates technologies like IoT, _____, and robotics into operations.	01	1	1
6	Processes can be categorized as _____ or supporting based on their direct impact on the final product.	01	1	1
7	Differentiate between order winner and order qualifier.	02	2	2
8	A hospital is considering a new procedure to be offered at Rs.200 per patient. The fixed cost per year would be Rs.100,000, with total variable costs of Rs.100 per patient. What is the break-even quantity for this service?	02	2	2
	Part –B			
1	Define operations management. Briefly explain the key terms involved in the definition of operations management with a schematic model of a production system.	10	2	1
2	With suitable example, Distinguish between Manufacturing and service processes.	10	3	1

Academic year 2024-2025 (ODD Sem)

3	White Valley Ski Resort is planning the ski lift operation for its new ski resort. Management is trying to determine whether one or two lifts will be necessary; each lift can accommodate 250 people per day. Skiing normally occurs in the 14-week period from December to April, during which the lift will operate 7 days per week. The first lift will operate at 90 percent capacity if economic conditions are bad, the probability of which is believed to be about a 0.3. During normal times the first lift will be utilized at 100 percent capacity, and the excess crowd will provide 50 percent utilization of the second lift. The probability of normal times is 0.5. Finally, if times are really good, the probability of which is 0.2, the utilization of the second lift will increase to 90 percent. The equivalent annual cost of installing a new lift, recognizing the time value of money and the lift's economic life, is $50,000. The annual cost of installing two lifts is only $90,000 if both are purchased at the same time. If used at all, each lift costs $200,000 to operate, no matter how low or high its utilization rate. Lift tickets cost $20 per customer per day. Should the resort purchase one lift or two?	20	4	3
4	A manager is deciding whether to build a small or a large facility. Much depends on the future demand that the facility must serve, and demand may be small or large. The manager knows with certainty the payoffs that will result under each alternative, shown in the following payoff table. What is the best alternative for each decision rule: Maximin. Maximax. Laplace. Minimax Regret.	10	4	3

	Possible future demand	
Alternative	Low	High
Small facility	200	270
Large facility	160	800
Do nothing	0	0

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DEPARTMENT OF
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Date : 14th Oct 2026	**CIE – 1**	**Max. Marks : 50**
Semester : V	**UG**	**Duration : 90 Min**
Course Title: Quality Assurance		**Course Code : IM353IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – B			
1.	Differentiate between the Cost of Good Quality and the Cost of Bad Quality in the automotive industry. Give examples of each and describe how managing these costs helps improve product quality and reduce waste.	10	3	2
2.	Evaluate the strategic, operational, and environmental benefits an organization gains from implementing the ISO 14000 standard, and explain how these contribute to long-term business sustainability.	10	4	3
3.	What is pattern analysis in control charts? Explain with an example.	10	3	3
4.	Analyze the role of the Eight Dimensions of Quality in determining customer acceptance of a selected product (e.g., smartphone, automobile, or home appliance). Evaluate how each dimension contributes to customer satisfaction and market competitiveness, and illustrate how these dimensions interact to strengthen the product's overall value proposition.	10	3	2
5.	A factory produces cylindrical rods. The length of rods is measured during production. Length Measurements (cm) Sample | Measurement 1 | Measurement 2 | Measurement 3 | Measurement 4 1 | 40.2 | 40.5 | 40.3 | 40.4 2 | 40.6 | 40.7 | 40.5 | 40.8 3 | 40.1 | 40.0 | 40.2 | 40.1 4 | 40.4 | 40.6 | 40.5 | 40.7 5 | 40.3 | 40.2 | 40.4 | 40.1 Given the standard factors for sample size $n = 4$: $A_2 = 0.729$, $D_3 = 0$, $D_4 = 2.282$ 1. Calculate the mean ($\bar{X}$) and range (R) for each sample. 2. Calculate the overall mean ($\bar{\bar{X}}$) and average range ($\bar{R}$). Using the given standard factors, determine: • $\bar{X}$-chart control limits: $UCL_{\bar{X}}, CL_{\bar{X}}, LCL_{\bar{X}}$ • R-chart control limits: UCL_R, CL_R, LCL_R 3. Interpret the $\bar{X}$-chart and R-chart. Determine whether the production process is in statistical control and identify any special-cause variations.	10	4	4

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Date : 7th Nov 2025	**CIE-1**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Quality Assurance		**Course Code : IM353IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List the statistical methods for quality control and improvement.	2	1	01
2.	What is the primary objective of ISO 14000 standards?	2	2	01
3.	Write the steps followed in constructing the individual chart	2	2	02
4.	Differentiate between defect and defective.	2	2	02
5.	What is the primary goal of quality assurance?	2	2	02
	Part – B			
1	"Analyze how Rational Subgrouping can be used as a quality control strategy in manufacturing or service industries to separate common and special causes of variation, thereby improving process stability."	10	4	3
2	"Evaluate the organizational, operational, and strategic benefits of implementing the ISO 9000 quality standard, and explain how it contributes to continuous improvement and competitive advantage."	10	3	3
3	"Analyze how the Eight Dimensions of Quality can be used to evaluate the performance of a consumer product. Examine each dimension's effect on customer satisfaction and illustrate your analysis with a detailed case study from the consumer goods or electronics industry."	10	4	3
4	A manufacturing company measures the diameter (in mm) of a component during production. Data are collected in	10	4	4

Sample	Measurement 1	Measurement 2	Measurement 3	Measurement 4
1	24.8	25.1	24.9	25.0
2	25.3	25.2	25.4	25.1
3	24.7	24.9	24.8	24.6
4	25.5	25.4	25.6	25.3
5	24.9	25.0	25.1	24.8
6	25.2	25.3	25.1	25.0

A) sample calculate the sample mean
B) determine the $\bar{X}$-chart control limits and the R-chart control limits:
C) Interpret the control charts: are the process means and ranges in control? Identify any points or patterns that indicate special-cause variation.

5	A process is producing shafts with a nominal diameter of 50 mm. The specification limits are 50 ± 0.5 mm. From control chart data, the process mean (X) is 50.1 mm, and the process standard deviation (σ) is 0.1 mm. (a) Calculate the Upper Control Limit (UCL) and Lower Control Limit (LCL) for the process using $\pm 3\sigma$ limits. (b) Compare the control limits with the specification limits and comment on the process capability.	10	4	4

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**26th November 2024**	**Maximum Marks**	**10 + 50**
Course Code	**IM3531A**	**Duration**	**120 Mins**
Sem	**V Semester**	**CIE – I**	
QUALITY ASSURANCE			

Note:
1. Answer all the Questions.

PART – A

Sl. No	Questions	M	BT	CO
1	Differentiate between defect and defective.	2	2	1
2	What is the primary goal of quality assurance?	1	2	1
3	How does ISO certification benefit organizations?	1	1	2
4	Mention four key core technologies that drive Quality 4.0.	2	2	3
5	List the 7 QC tools	2	1	3
6	What is ATS and ARL?	2	1	2

PART – B

Sl. No	Questions	M	BT	CO
1.	A certain type of storage of battery lasts on an Average 3 years and Standard of 0.5, assume normal distribution, find percentage of battery that lasts a) Less than 2 years, b) Less than 4 years, c)between 2.5 to 4.5 years, d)less than 4 years.	10	3	3
2.	Describe the principles of Rational Subgrouping and apply them to a specific industry, such as manufacturing or service, to show how subgrouping can be used to reduce variation and improve process quality.	10	2	1
3.	Classify the Cost of Good Quality and Cost of Bad Quality in the automotive industry. Provide examples of each cost category and explain how they can be managed effectively to improve product performance and reduce overall expenses.	10	1	2
4.	Choose a specific product (e.g., a smartphone, automobile, or any consumer electronics) and analyze how the Eight Dimensions of Quality influence its acceptance by customers. Break down each dimension—such as performance, features, reliability, and serviceability—and evaluate their relative importance in shaping customer perceptions. How do these dimensions interact with each other to create a competitive advantage for the product in the market? Use relevant examples to support your analysis.	10	4	4
5.	What are the key benefits of implementing the ISO 14000 standard for an organization?	10	2	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

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	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	Max Marks	Quiz	03	06	04	–	–	05	05	–	–	–	–
		Test	10	20	10	10	–	10	20	10	10	–	–

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 13th Oct 2026	**CIE – I**	**Max. Marks : 50**
Semester : V	**UG**	**Duration : 90 Min**
Course Title: FINANCIAL ACCOUNTING & COSTING		**Course Code : IM254TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
1.	Explain how cost accounting information can help engineers make "make or buy" decisions in production planning using an example.	10	2	1
2.	Pass the necessary journal entries in the book of Raghav from the following transactions: (a) On 1 April 2023, opened a Bank Account by depositing ₹50,000. (b) On 7 April 2023, goods were brought for ₹10,000, and payment was made by cheque. (c) On 9 April 2023, Ravi (the debtor) directly deposited an amount of ₹40,000 in Raghav's account. (d) On 15 April 2023, cash withdrawn ₹5,000 for personal use. (e) On 21 April 2023, sold goods to Ankit for ₹30,000 at a cash discount of 10% and received a cheque for the full amount deposited into the bank the same day. (f) On 26 April 2023, the cheque received from Ankit was dishonoured. (g) On 29 April 2023, the Bank charged interest for ₹500.	10	3	2
3.	What is the significance of a trial balance? Explain its usefulness and limitations in the preparation of financial statements.	10	3	3
4.	For the following transactions, prepare the ledger balance for cash account and purchase account. 1. Purchased goods for cash – Rs 5000 2. Sold goods for cash – Rs 8000 3. Received cash from Deepak – Rs 2500 4. Paid rent – Rs 5000 5. Deposited in bank – Rs 40000 6. Sold goods to Hari – Rs 2500	10	3	3
5.	Differentiate between **fixed** and **flexible** budgets. Under what situations would each be used in a manufacturing organization?	10	3	3

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 07/11/2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: FINANCIAL ACCOUNTING & COSTING		**Course Code : IM254TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	State the primary objective of financial accounting	1	1	1
2.	The process of recording a transaction is called _____.	1	2	2
3.	Showcase your understanding of accounting equation with an example	2	2	2
4.	Identify the books which are termed as “First book of accounting and Principal book of accounting”?	2	2	1
5.	State any two advantages of adopting IFRS for Indian engineering firms with global operations.	2	2	3
6.	In the era of many startups trying to be unicorns, comment on the “Conservatism/Prudence concept” and its importance which has helped traditional businesses prevail in the long run.	2	2	2
	Part – B			
1.	Review the Golden Rules of Accounting. Illustrate each category with necessary examples.	10	2	1
2.	Assume you are starting an E-Commerce website to sell a product of your choice. Identify and list any 10 financial transactions that you might encounter in the first month of starting the business and pass them on to the journal books.	10	4	2
3.	The following is a list of various accounts. State the nature of each account (personal, real or nominal) and also which account will be debited and which account will be credited. (**Use a tabular column to furnish your answer.**) Wages outstanding, Capital introduced, Cartage paid, Prepaid salary, Rent outstanding, Bank overdraft, Discount allowed, Goods returned by buyer, Drawings by proprietor, Commission received.	10	3	3

<table>
<tr><td>4.</td><td>Give the Journal entries for the following transactions:
<table>
<tr><td>Date</td><td>Particulars</td><td>₹</td></tr>
<tr><td>April 1</td><td>Business started with cash</td><td>10,000</td></tr>
<tr><td>April 3</td><td>Deposited in SBI</td><td>6,000</td></tr>
<tr><td>April 5</td><td>Bought goods from Mahavir</td><td>1,500</td></tr>
<tr><td>April 9</td><td>Sold goods to Gupta</td><td>650</td></tr>
<tr><td>April 12</td><td>Paid cash to Mahavir</td><td>990</td></tr>
<tr><td>April 15</td><td>Cash received from Gupta</td><td>625</td></tr>
<tr><td>April 20</td><td>Furniture purchased</td><td>300</td></tr>
<tr><td>April 22</td><td>Drew cash from bank for personal use</td><td>600</td></tr>
<tr><td>April 30</td><td>Paid rent by cheque</td><td>200</td></tr>
<tr><td>April 30</td><td>Salary due to clerk</td><td>300</td></tr>
</table>
Also prepare the necessary ledger accounts for cash and capital accounts only.</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>5.</td><td>Financial literacy is very important for everyone given the road map India has chosen to grow. There are several challenges on the ground also with regard to providing “Financial Literacy”.
Conceptualize a framework which utilizes students with both engineering & finance knowledge along with problem solving skills like you to provide “Financial knowledge” to students between the age group of 15 to 19. You are allowed to use the same flow which you used for your “Design Thinking Project”. Comment on Novelty that you want to bring from your POV.</td><td>10</td><td>4</td><td>3</td></tr>
</table>

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**26th November 2024**	**Maximum Marks**	**10 + 50**
Course Code	**IM254TA**	**Duration**	**120 Mins**
Sem	**V Semester**	**CIE – I**	
FINANCIAL ACCOUNTING AND COSTING			

Note:
1. Answer all the Questions.

PART – A

Sl. No	Questions	M	BT	CO
1	Write a note on double-entry bookkeeping.	02	2	1
2	List the main financial statements prepared in financial accounting	02	1	1
3	Mention the accounting equation	02	1	1
4	What is the purpose of a trial balance?	02	2	1
5	Depict the journal format in accounting.	02	1	1

PART – B

Sl. No	Questions	M	BT	CO
1.	Explain the steps involved in the accounting cycle and discuss their significance in ensuring accurate financial reporting	10	3	1
2.	Discuss how zero-based budgeting differs from traditional budgeting approaches. What are the benefits and limitations of adopting a zero-based budgeting system?	10	2	1
3 a.	Distinguish between Single Entry vs. Double Entry accounting process.	05	2	1
b.	Write a note on International Financial Reporting Standards.	05	2	1
4 a.	Journalise the business transactions during the month of Jan 2024. Jan 1: Owner invested ₹50,000 cash in the business. Jan 3: Purchased office furniture for ₹10,000, paid in cash. Jan 5: Bought goods worth ₹20,000 on credit from M/s XYZ Traders. Jan 10: Paid ₹5,000 to M/s XYZ Traders as partial payment. Jan 15: Sold goods for ₹15,000 on cash. Jan 20: Paid office rent of ₹3,000 in cash.	05	3	2
b.	Discuss the principles of debit and credit in journalizing transactions. How do these principles apply to different types of accounts?	05	3	1

5.	Journalise the business transactions during the month of Feb XXXX Feb 2: Started business with cash ₹1,00,000 and machinery ₹50,000. Feb 5: Opened a bank account by depositing ₹30,000. Feb 10: Purchased goods for ₹40,000 on credit from M/s ABC Traders. Feb 12: Paid ₹5,000 for office rent in cash. Feb 17: Sold goods for ₹25,000 on credit to M/s XYZ Ltd. Feb 20: Paid ₹10,000 to M/s ABC Traders. Feb 22: Purchased office supplies for ₹2,000 in cash. Feb 24: Received ₹20,000 from M/s XYZ Ltd. Feb 27: Paid wages ₹4,000 in cash. Feb 28: Withdrew ₹3,000 for personal use.	10	3	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5
Marks Distribution	Max Marks	Quiz	10	–	–	–	–	06	04	–	–	–
		Test	35	15	–	–	–	–	20	30	–	–

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